Workshop 7

Math 292 Spring 2026
Rutgers University

Exercise 1: Let $[a, b]$ be a bounded interval. Suppose $p(x)$ is strictly positive and continuously differentiable on $[a, b]$. Consider

$$\mathcal{L}u(x) = (p(x)u'(x))'$$

and the eigenfunctions $\mathcal{L}u(x) = \lambda u(x)$ with Neumann boundary conditions

$$u'(a) = u'(b) = 0.$$

Multiplying both sides of $\mathcal{L}u(x) = \lambda u(x)$ by $u(x)$ and integrating, we have

$$\begin{aligned}
\int_a^b \mathcal{L}u(x) \cdot u(x)\, dx &= \int_a^b (p(x)u'(x))' \cdot u(x)\, dx \\
&= \Big[p(x)u'(x)u(x)\Big]_a^b - \int_a^b p(x)(u'(x))^2\, dx \\
&= -\int_a^b p(x)(u'(x))^2\, dx \\
&= \int_a^b \lambda |u(x)|^2\, dx.
\end{aligned}$$

That is,

$$\lambda = -\int_a^b \frac{p(x)(u'(x))^2}{|u(x)|^2}\, dx.$$

Since $p(x) > 0$, $u(x)$ is nontrivial, and the square of reals are non-negative, it follows that $\lambda \leq 0$.

Exercise 2: Let u_λ be an nontrivial eigenfunction for $\mathcal{L}$ with $\lambda \neq 0$, and let x_0 be any critical point on $[A, B]$.

For the sake of contradiction, suppose $u_\lambda(x_0) = 0$. Since x_0 is a critical point, it follows that $u'(x_0) = 0$.

Then, we can rewrite the eigenfunction $\mathcal{L}u(x) = \lambda u(x)$ as

$$u''(x) + \frac{p'(x)}{p(x)}u'(x) - \frac{\lambda}{p(x)}u(x) = 0. \tag{1}$$

Notice that this is a second order linear ODE. So by Picard-Linedelof's Theorem, there must exist a unique solution for each initial condition.

Consider the constant function $v(x) = 0$. Notice it satisfies our initial conditions $u(x) = u'(x) = 0$, and thus it is a valid solution. Moreover since $u_\lambda(x_0)$ also satisfies $u(x) = u'(x) = 0$, by uniqueness of solutions, they must be the same. However, this is impossible since $v(x)$ is the trivial solution and we claimed that $u_\lambda(x_0)$ is a nontrivial solution. Thus, we have a contradiction. Therefore, $u_\lambda(x_0) \neq 0$.

Consider equation 1 again. Evaluating at x_0:

$$u''(x_0) + \frac{p'(x_0)}{p(x_0)}u'(x_0) - \frac{\lambda}{p(x_0)}u(x_0) = u''(x_0) - \frac{\lambda}{p(x_0)}u(x_0) = 0.$$

Then,

$$u''(x_0) = \frac{\lambda}{p(x_0)}u(x_0).$$

From exercise 1, we have that $\lambda \leq 0$ and we let $\lambda \neq 0$, so $\lambda < 0$ must be true. Since $\lambda < 0$ and $p(x_0) > 0$, it follows that $\frac{\lambda}{p(x_0)} < 0$. Then, $u''(x_0)$ is $u(x)$ times a negative scalar, and so they must have opposite signs.

Moreover, since we have $u_\lambda(x_0) \neq 0$, $\lambda < 0$, and $p(x_0) > 0$, it follows that $u''(x_0) \neq 0$.

Exercise 3: Let $(x_1, x_2) \in [A, B]$ be successive critical points of u_λ. Then,

$$u'_\lambda(x_1) = u'_\lambda(x_2) = 0.$$

Define $v_\lambda := u_\lambda|_{[x_1,x_2]}$. Then $\mathcal{L}v_\lambda(x) = \lambda v_\lambda(x)$ still holds because $[x_1, x_2] \subset [A, B]$. Since x_1 and x_2 are critical points, $v'_\lambda(x_1) = 0$ and $v'_\lambda(x_2) = 0$ are both true. So Neumann boundary conditions are also satisfied.

Now consider the constant function $w(x) = 1 : \forall x \in [x_1, x_2]$. Notice that $w(x)$ is an eigenfunction of $\mathcal{L}$ with $\lambda = 0$ since

$$\mathcal{L}w(x) = (p(x) \cdot w'(x))' = 0 = 0 \cdot w(x).$$

It also satisfies the Neumann boundary conditions:

$$w'(x_1) = w'(x_2) = 0.$$

Recall that eigenfunctions with different eigenvalues must be orthogonal to each other. Then, since we have $\lambda \neq 0$:

$$\int_{x_1}^{x_2} v_\lambda(x) \cdot w(x)\ dx = \int_{x_1}^{x_2} v_\lambda(x)\ dx = 0.$$

Now suppose, for sake of contradiction, that $v_\lambda(x) \neq 0 : \forall x \in (x_1, x_2)$. From Exercise 2, we also have that $v_\lambda(x_1), v_\lambda(x_2) \neq 0$. So our bounds expand to

$$v_\lambda(x) \neq 0 : \forall x \in [x_1, x_2].$$

Then $v_\lambda(x)$ is a non-zero continuous function on a bounded interval. So by Intermediate Value Theorem, $v_\lambda(x)$ never switch signs, and thus we have these two cases.

Case 1: We have $v_\lambda(x) > 0 : \forall x \in [x_1, x_2]$. Then,

$$\int_{x_1}^{x_2} v_\lambda(x) > 0.$$

This contradicts our earlier findings that $\int_{x_1}^{x_2} v_\lambda(x) = 0$.

Case 2: We have $v_\lambda(x) < 0 : \forall x \in [x_1, x_2]$. Then,

$$\int_{x_1}^{x_2} v_\lambda(x) < 0.$$

This again contradicts our earlier findings that $\int_{x_1}^{x_2} v_\lambda(x) = 0$.

Since both cases lead to contradictions, it must be that our assumption is false. Therefore $v_\lambda(x)$ has at least one zero in the interval (x_1, x_2).

Exercise 4: Set $\mathcal{L}u(x) = ((1+x)u'(x))'$, so that $V_{\lambda_2}(x) = \frac{1}{4}\frac{1}{(1+x)^2} - \frac{\lambda_2}{1+x}$ for Dirichlet boundary conditions on $[x_1, x_2]$. Assume $x_1, x_2 > 0$. Differentiating $V_{\lambda_2}(x)$, we have

$$V'_{\lambda_2}(x) = -\frac{1}{2}(1+x)^{-3} + \lambda_2(1+x)^{-2}.$$

Since $x_1, x_2 > 0$, $x > 0$ in this interval. Moreover, we have that $\lambda_2 < 0$ from the previous exercises. So $V'_{\lambda_2}(x)$ must be strictly negative for all $x \in [x_1, x_2]$. That is, $V_{\lambda_2}(x)$ is monotonically decreasing for all $x \in [x_1, x_2]$. Then, $V_{\lambda_2}(x)$ attains it's maximum at x_1 and minimum at x_2. Specifically,

$$V_{\lambda_2}(x_1) = \frac{1}{4}\frac{1}{(1+x_1)^2} - \frac{\lambda_2}{1+x_1} = M^2,$$

and

$$V_{\lambda_2}(x_2) = \frac{1}{4}\frac{1}{(1+x_2)^2} - \frac{\lambda_2}{1+x_2} = m^2.$$

Recall that $y(x) = \sqrt{p(x)} \cdot u_{\lambda_2}(x)$ where $p(x) > 0$ and $u_{\lambda_2}(x)$ is an nontrivial eigenfunction. So $y(x)$ is a non-trivial solution. Since we also have that x_1 and x_2 are successive zeroes, by Theorem 2.1, we have

$$\frac{\pi}{M} \leq x_2 - x_1 \leq \frac{\pi}{m}.$$

We can rearrange this to

$$m \leq \frac{\pi}{x_2 - x_1} \leq M.$$

Squaring everything, we have

$$m^2 \leq \frac{\pi^2}{(x_2 - x_1)^2} \leq M^2.$$

Using our substitutions for M^2 and m^2, we solve for λ_2.

For the upper bound:

$$\begin{aligned} M^2 &\geq \frac{\pi^2}{(x_2 - x_1)^2} \\ \frac{1}{4(1+x_1)^2} - \frac{\lambda_2}{1+x_1} &\geq \frac{\pi^2}{(x_2 - x_1)^2} \\ \frac{\lambda_2}{1+x_1} &\leq \frac{1}{4(1+x_1)^2} - \frac{\pi^2}{(x_2 - x_1)^2} \\ \lambda_2 &\leq \frac{1}{4(1+x_1)} - \frac{\pi^2(1+x_1)}{(x_2 - x_1)^2} \end{aligned}$$

For the lower bound:

$$\begin{aligned} m^2 &\leq \frac{\pi^2}{(x_2 - x_1)^2} \\ \frac{1}{4(1+x_2)^2} - \frac{\lambda_2}{1+x_2} &\leq \frac{\pi^2}{(x_2 - x_1)^2} \\ \frac{\lambda_2}{1+x_2} &\geq \frac{1}{4(1+x_2)^2} - \frac{\pi^2}{(x_2 - x_1)^2} \\ \lambda_2 &\geq \frac{1}{4(1+x_2)} - \frac{\pi^2(1+x_2)}{(x_2 - x_1)^2} \end{aligned}$$

Combining, we have

$$\frac{1}{4(1+x_2)} - \frac{\pi^2(1+x_2)}{(x_2 - x_1)^2} \leq \lambda_2 \leq \frac{1}{4(1+x_1)} - \frac{\pi^2(1+x_1)}{(x_2 - x_1)^2}.$$